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Bistability and Correlation with Arrhythmogenesis in a Model of the Right Atrium

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Abstract—Rapid pacing is an important tool for understanding cardiac arrhythmias. A recent experiment involving rapid pacing of sheep atria indicated that the initiation of atrial arrhythmias may be related to the 1:1/2:1 bistability. To elucidate the mechanism of this relation, this study applied the pacing protocol from the sheep study to an idealized model of the right atrium. The model included all major anatomical features, the sino-atrial node, and the regional differences in the action potential duration (APD). A pacing protocol was applied, in which the basic cycle length (BCL) was decreased in steps of 10 ms until the response switched to 2:1, then BCL was increased. The 1:1-to-2:1 transitions occurred at shorter BCLs than the 2:1-to-1:1 transitions yielding a global bistability window of 60 ms. As in the sheep study, idiopathic waves were observed at BCLs within or near the bistability window. The model was used to quantify the types, prevalence, and persistence of idiopatic waves, study their initiation and termination, and relate them to the model components. The results demonstrate that idiopatic waveforms move with the shift of the bistability window and that they disappear when bistability is eliminated. Thus, this modeling study supports causal relationship between the 1:1/2:1 bistability and the initiation of arrhythmias.

Keywords—Atrial arrhythmias, Atrial model, Cardiac dynamics, Numerical simulations.

INTRODUCTION

A revealing method used to investigate the electrical dynamics of cardiac tissue is periodic pacing at different basic cycle lengths (BCLs). Numerous studies have investigated the responses of cardiac tissue as a function of BCL.[6–9,12,13,28] The observed responses included various phase-locked M:N responses where M is the number of stimuli and N is the number of beats, as well as irregular responses, where beats are not correlated with the stimuli. Two different phase-locked or irregular responses can occur at the same BCL, a phenomenon known as bistability. A commonly observed bistability is 1:1/2:1 bistability, first observed by Mines in 1913 in frog ventricles.[12] Since then, 1:1/2:1 bistability has been observed in a variety of preparations, from single cells to *in vivo* hearts,[6–9,13,28] suggesting that it is a characteristic common to cardiac tissue. In addition, a study performed on sheep atria *in vivo*[16] observed that idiopathic waves (waves of unknown origin not resulting from the stimuli) occurred at BCLs within the bistability window or near its limits. The idiopathic waves were similar to those in various atrial arrhythmias suggesting a link between 1:1/2:1 bistability and arrhythmogenesis.

However, the sheep study had two important limitations: 1) membrane potential, the most revealing measure of cardiac tissue behavior, was not measured, and 2) only a small portion of the right atrium was mapped, which made it impossible to determine how the tissue behaved away from the mapping plaque and to identify the mechanisms responsible for idiopathic wave initiation, maintenance, and termination. The work presented here supplements previous experimental study by using computer simulation of pacing and wave propagation in an idealized model of the right atrium. The model was used to further explore the link between 1:1/2:1 bistability and idiopathic waves and to explain the role of anatomic structures, electrophysiologic heterogeneity, and local and global bistability in the initiation and termination of idiopathic waves during programmed pacing.

METHODS

Geometric Model

The geometry of the atrium was idealized as a two-dimensional sheet, 12.5 × 10 cm, converted to a cylinder by applying periodic boundary conditions to the sides of the sheet and no-flux boundary conditions to the top and bottom of the sheet. The cylinder was large enough to sustain circumferential and spiral-wave reentry. The top and bottom of the cylinder provided two anatomic openings that roughly approximated the superior vena cava (SVC) and inferior vena cava (IVC). The relative size and location of the other anatomic structures [coronary

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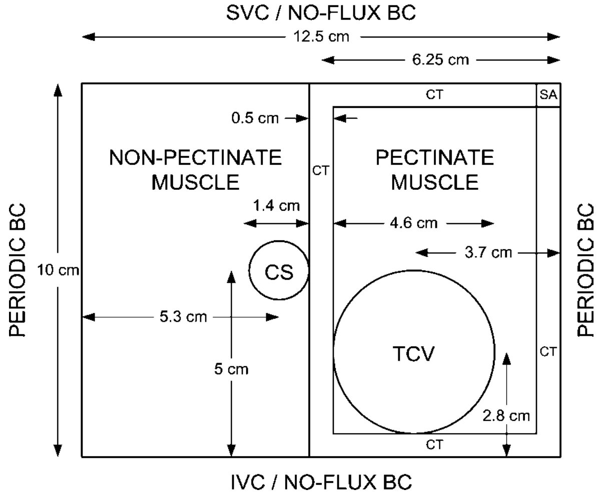

FIGURE 1. Idealized model of the right atrium. PERIODIC BC = periodic boundary condition. NO-FLUX BC = no-flux boundary condition. SVC = superior vena cava. IVC = inferior vena cava. CS = coronary sinus (diam = 1.4 cm). TCV = tricuspid valve (diam = 4.6 cm). CT = crista terminalis (width = 0.5 cm). SA = sino-atrial node (0.5×0.5 cm).

sinus (CS), tricuspid valve (TCV), crista terminalis (CT), pectinate muscle, non-pectinate muscle, and sinoatrial (SA) node] were chosen based on echocardiograms[2] and dissections[1] of human hearts, an anatomically correct model (Bobbitt Laboratories; Burlington, North Carolina), and drawings[15] of the human heart. Note that the sections of the CT that run vertically and border the pectinate muscle on the left and right sides will hereafter be referred to as the left CT (LCT) and right CT (RCT), respectively. All details of the completed model are shown in Fig. 1.

Membrane Model

A preferred approach to reproduce the results of the experimental study on sheep atria would be to use a physiologically accurate model of the atrial membrane, e.g., the Courtemanche-Ramirez-Nattel (CRN) model.[3] However, such models are computationally demanding, particularly for the long pacing protocols required for this study. Therefore, a three-variable membrane model, derived from the Fenton-Karma cardiac membrane model[5] was used. This model has been shown to reproduce many of the dynamic features observed in more complex, ionic-based models. The model parameters were adjusted to mimic the action-potential-duration (APD) restitution and conduction-velocity (CV) restitution of the CRN model since these two relationships are fundamental to modeling cardiac dynamics.[5] The details of choosing the parameters are reported elsewhere.[17] The resulting FKCRN model and the parameter values used are shown in Table 1. The notation used in Table 1 is that used by our group; specifically, most variables were renamed to reflect their function.

Experiments have shown that APD varies spatially across the right atrium[11,14,23] reflecting the underlying electrophysiologic heterogeneity. A review of the limited and varied data from different species and experimental preparations[4,10,18,22,27] suggests that APD is longer in the CT than in the non-pectinate muscle and longer in the non-pectinate muscle than in the pectinate muscle. Using those data as a guide, this study assumed that $\mathrm{APD_{CT}} = 1.40\,\mathrm{APD_{non-pect}}$ and $\mathrm{APD_{non-pect}} = 1.05\,\mathrm{APD_{pect}}$. In the model, this APD heterogeneity was achieved by varying τ_{ung2}, a time constant which scales the ungated outward current of the FKCRN model. Specifically, τ_{ung2} and intrinsic APD were 601.0 ms and 469.6 ms, respectively,

TABLE 1. Fenton-Karma membrane model.

Potential equation: $\dfrac{dv}{dt} = -(J_{\mathrm{fast}} + J_{\mathrm{slow}} + J_{\mathrm{ung}} - J_{\mathrm{stim}})$

Fast inward current: $J_{\mathrm{fast}} = -\frac{f}{\tau_{\mathrm{fast}}} F(v)$; $v < V_f$: $F(v) = 0$; for $v > V_f$: $F(v) = (v - V_f)(1 - v)$

Fast gate equation: $\dfrac{df}{dt} = \dfrac{f_\infty - f}{\tau_f}$

$v < V_f$: $f_\infty = 1, \tau_f = \tau_{\mathrm{fopen}}$; $v < V_{\mathrm{fopen}}$: $\tau_{\mathrm{fopen}} = \tau_{\mathrm{fopen1}}$; $v > V_{\mathrm{fopen}}$: $\tau_{\mathrm{fopen}} = \tau_{\mathrm{fopen2}}$

$v > V_f$: $f_\infty = 0, \tau_f = \tau_{f\,close}$

Slow inward current: $J_{\mathrm{slow}} = -\frac{s}{\tau_{\mathrm{slow}}} S(v)$; $S(v) = \frac{1}{2}(1 + \tanh(k(v - V_{\mathrm{slow}})))$

Slow gate equation: $\dfrac{ds}{dt} = \dfrac{s_\infty - s}{\tau_s}$; $v < V_s : s_\infty = 1, \tau_s = \tau_{\mathrm{sopen}}$; $v > V_s : s_\infty = 0, \tau_s = \tau_{\mathrm{sclose}}$

Ungated outward current: $J_{\mathrm{ung}} = \frac{1}{\tau_{\mathrm{ung}}} U(v)$

$v < V_u : U(v) = v, \tau_{\mathrm{ung}} = \tau_{\mathrm{ung1}}$

$v > V_u : U(v) = 1, \tau_{\mathrm{ung}} = \tau_{\mathrm{ung2}}$

Parameter values chosen to reproduce APD and CV restitution of CRN atrial model:

$V_f = V_s = V_u = 0.160$, $V_{\mathrm{fopen}} = 0.040$, $V_{\mathrm{slow}} = 0.85$, $k = 10.0$

$\tau_{\mathrm{fast}} = 0.249$ ms, $\tau_{\mathrm{fopen1}} = 40.0$ ms, $\tau_{\mathrm{fopen2}} = 82.5$ ms, $\tau_{\mathrm{fclose}} = 5.75$ ms

$\tau_{\mathrm{slow}} = 226.9$ ms, $\tau_{\mathrm{sopen}} = 100.0$ ms, $\tau_{\mathrm{sclose}} = 300.0$ ms, $\tau_{\mathrm{ung1}} = 64.7$ ms, $\tau_{\mathrm{ung2}} = 222.9$ ms

Converting dimensionless v to membrane potential V_m:

$V_m = V_{\mathrm{rest}} + v(V_{\mathrm{peak}} - V_{\mathrm{rest}})$; $V_{\mathrm{rest}} = -81.2$ mV, $V_{\mathrm{peak}} = 3.6$ mV

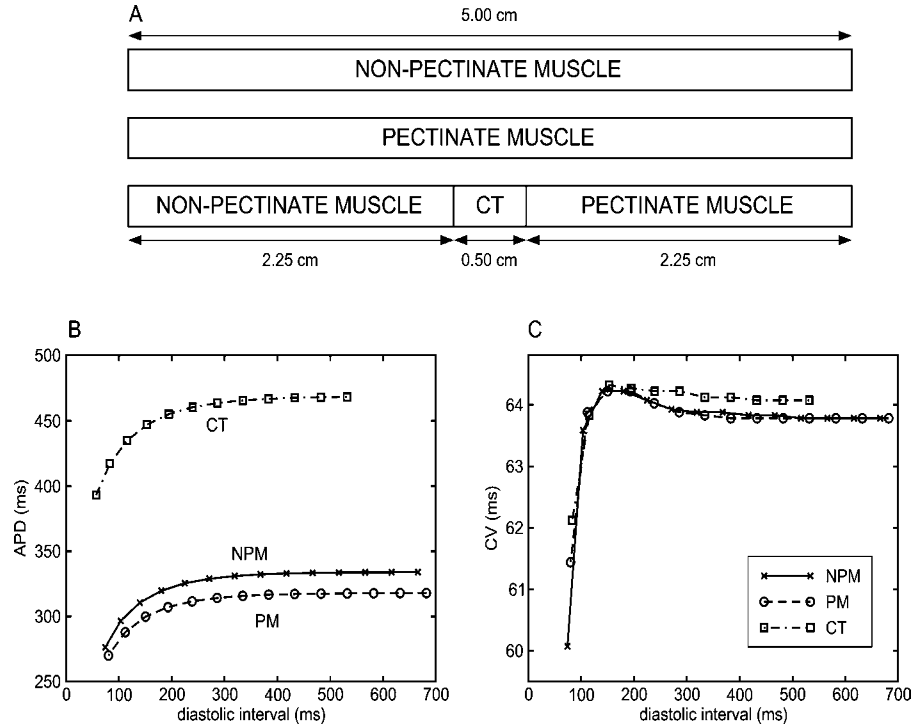

FIGURE 2. (A) Fibers excised from the model and used to determine local restitution properties and bifurcation diagrams of the underlying regions. Stimulus site at far left end and measurement site in the center. Steady-state APD (panel B) and CV (panel C) restitution curves for the three fibers. NPM = non-pectinate muscle (*x*). PM = pectinate muscle (circles). CT = crista terminalis (squares).

in the CT; 222.9 ms and 335.2 ms in the non-pectinate muscle; and 213.9 ms and 319.8 ms in the pectinate muscle.

To assess the local restitution properties of different regions, three 2-cm fibers were excised from the atrium model as shown in Fig. 2A. The membrane of the first and second fiber had the properties of the non-pectinate and pectinate muscle, respectively. The third fiber had three sections: a 0.2 cm section representing the CT model in the middle surrounded by regions representing non-pectinate and pectinate muscle. Such a heterogeneous fiber was chosen to reproduce the electrotonic coupling that affects APD of the CT in the atrium model. For each fiber model, stimuli were applied at the left end of the fiber and APD was measured in the center. The resulting steady-state APD and CV restitution curves are shown in Fig. 2B-C. The APD restitution curves are monotonic; the CV restitution curves show supernormal conductivity, a feature also present in the CRN model.[17]

SA Node

The model also included an SA node region that was self-excitatory at a constant activation cycle length (ACL), capable of launching activation waves that activated the model, and resettable such that if activated externally subsequent self-excitation was delayed. Spontaneous activation in the SA node is caused by an inward current that depolarizes the membrane potential. Thus, a simple pacemaker current, J_{pace}, was added to the FKCRN model. Specifically, when membrane potential recovered to a small value, $V_{\text{depol}} = 0.0042$, the pacemaker current would activate, $J_{\text{pace}} = -v/\tau_{\text{pace}}$, where $\tau_{\text{pace}} = 10^{-4}$ ms. The resulting spike of inward current would depolarize the tissue. The SA node is located in the right atrium adjacent to the SVC in the CT. In the model it occupied a 0.5×0.5 cm region as shown in Fig. 1.

Moving and Eliminating Bistability Window

To investigate whether there is a causal relationship between bistability and idiopathic waves, the global bistability window was first shifted in BCL and then eliminated altogether. To shift the window, a time constant τ_{slow} that scales the slow inward current J_{slow} was changed. The nominal case had the parameter values given in Table 1. To move the bistability window to longer (shorter) BCLs, hereafter referred to as the long (short) case, τ_{slow} was decreased (increased) throughout the model. Values of τ_{slow} for all three cases are listed in Table 2. To eliminate bistability window, the short case was modified by reducing τ_{ung1}, a time constant that scales the ungated outward current, J_{ung} (Table 2). τ_{ung1} was chosen because further increase in τ_{slow} beyond that of the short case no longer had any appreciable

TABLE 2. Results of downsweep/upsweep protocol.

Case	τ_{slow}	τ_{ung1}	SA ACL	1:1-to-2:1	2:1-to-1:1	Window
Long	183.1	64.7	752	455	395	60
Nominal	226.9	64.7	660	365	305	60
Short	606.9	64.7	550	335	275	60
None	606.9	49.7	500	320	320	0

Note. τ_{slow} (τ_{ung1}): parameter value of the FKCRN model that shifted (eliminated) global bistability window. SA ACL: activation cycle length of the SA node. 1:1-to-2:1: BCL at which 1:1-to-2:1 transition occurs. 2:1-to-1:1: BCL at which 2:1-to-1:1 transition occurs. Window: width of global bistability window. Long, nominal, short: cases with global bistability window shifted. None: case with global bistability window eliminated. (all values are in ms)

effect on the bistability window, and τ_{ung2}, the second time constant of J_{ung} had been used to effect APD heterogeneity across the model.

Governing Equation and Numerical Solution

Electrical activation and propagation were simulated using the monodomain model. Intracellular resistivity was considered homogeneous (not a function of position) and isotropic (not a function of direction) and therefore was constant. The governing equation was:

$$\frac{1}{R_i}\nabla^2 V_m = \beta\left(C_m\frac{\partial V_m}{\partial t} + I_{ion} - I_{stim}\right) \tag{1}$$

where V_m is membrane potential in mV, R_i is intracellular resistivity (320 Ω cm), β is surface to volume ratio (2500 cm^{-1}), C_m is membrane surface capacitance[3] (1.99 μF/cm^2), t is time in ms, I_{ion} is ionic current in μA/cm^2 computed using the FKCRN model shown in Table 1, and I_{stim} is stimulus current (128.8 μA/cm^2 for 4.5 ms, which is 1.25 times diastolic threshold). The chosen value of R_i results in CV at long BCLs of approximately 64 cm/s (Fig. 2C), which is typical for human atria.[20]

Equation 1 was solved numerically using a semi-implicit, finite-difference method based on a modified Crank-Nicolson algorithm,[19] and the gating variables were computed using an Euler method. The sheet was discretized using a regular grid with the same internodal spacing in the x and y directions, 250 μm. Time was discretized by using a constant time step of 120 μs. The spatial step and the time step were conservatively chosen such that APD restitution, CV restitution and action potential shape changed by less than 5% when each step was halved. All computer simulations discussed hereafter were performed using the CardioWave cardiac simulation package,[19] which uses the C programming language. All data input to CardioWave was generated using original programs written in Matlab (The MathWorks, Natick, MA, USA). All data output from CardioWave was post-processed using original programs written in Matlab. Because this study involved pacing protocols lasting several minutes in real time, simulations were performed on 16 IBM SP processors in parallel (International Business Machines Corporation, Armonk, NY, USA) to reduce overall computation time approximately 10 times. These simulations were run at the North Carolina Supercomputing Center (Research Triangle Park, NC, USA).

Pacing Protocols

Pacing stimuli were applied to one node in the pectinate muscle above the TCV which was approximately the position of the stimulus electrode during the sheep experiments,[16] i.e., the free wall of the right atrium. This site is marked with an asterisk in all activation maps involving pacing. Downsweep/upsweep protocols are described by four numbers, e.g., 450/250/400/10 means that pacing started at BCL of 450 ms, was decreased in steps of 10 ms to 250 ms, then increased in steps of 10 to 400 ms. If only a downsweep were applied, it would be denoted as 450/250/10.

In all downsweeps, the shortest BCL was selected based on a preliminary downsweep that identified at what BCL the 1:1-to-2:1 response transition occurred. The downsweep was immediately followed by a downsweep/upsweep which was applied to examine in more detail the range of BCLs around the 1:1-to-2:1 and 2:1-to-1:1 transitions. Step change in BCL was 50 ms in downsweeps and 10 ms in downsweep/upsweeps which assured manageable computation time and data storage while maintaining sufficient resolution in BCL. Like in the sheep experiment, eight paces were delivered at each BCL. Because FKCRN has only a limited memory,[24] this is sufficient to bring APD near steady state and determine the response type, while keeping computation time within reasonable limits. Eight paces may not be sufficient if a BCL falls very close to a bifurcation point, where transients are much longer. Such a case never occurred in our simulations but it is a limitation of this pacing protocol that can potentially affect the response type and shift the transition points.

Data Collection and Analysis

The full time course of membrane potential throughout pacing was determined at several locations in the model, including the stimulus site and two locations each within the non-pectinate muscle and pectinate muscle, and one location each within the CT and SA node. The electrical activity of the entire model was characterized by computing activation time and APD at 8000 uniformly distributed positions. The activation time was taken as the time the membrane potential crossed −40 mV from below during the upstroke. APD was the time interval between the activation time and 91.3% recovery. This recovery percentage minimized the effect of the stimulus artifact on APD when it was computed near the pacing site, but the resulting APD values were longer than those reported in the literature.[3]

Activation times were used to display activation patterns and create activation maps. Each activation pattern of the entire model was analyzed and classified as 1:1, 2:1, idiopathic waves, or some combination thereof. A global 1:1 response was defined as a wave moving radially outward from the stimulus site following each stimulus. A global 2:1 response was defined as such a wave following every other stimulus. Idiopathic waves were defined as those that did not originate at the stimulus site and were not synchronized with the stimuli.

To relate global response to the underlying tissue dynamics, in some simulations the local response at several nodes was determined. The local response was classified as 1:1, 2:1, 2:2 or irregular, based on the APD series computed at this node. Throughout this article, "local response" and "local bistability" refers to individual structures of the model while "global response" and "global bistability" refers to the model as a whole.

RESULTS

Nominal Case

Without external pacing, the SA node activated periodically and sent out waves that activated the entire model (Fig. 3, top). The activation pattern repeated with an ACL of 660 ms. When a 1000/250/50 downsweep was applied, both the SA node and the stimulus competed for control of the model. As BCL decreased during the downsweep, the impact of the stimulus on the global activity increased by: 1) causing wave breaks, 2) inducing idiopathic waves, and 3) intermittently overdriving the SA node. At a BCL of 550 ms, the stimulus had overtaken the SA node resulting in a 1:1 response with a stimulus wave moving radially outward from the stimulus site (Fig. 3, bottom).

The final conditions at a BCL of 450 ms on the downsweep were used as initial conditions for a 450/250/450/10 downsweep/upsweep that was applied to the model.

As BCL decreased from 450 to 390 ms, the response remained 1:1 with activation patterns similar to those of the bottom panel of Fig. 3. At a BCL of 380 ms the response switched from 1:1 to idiopathic waves as illustrated in Panel A of Fig. 4. A stimulus wave initially moved outward from the stimulus site as in a normal 1:1 response. However, instead of crossing the LCT, the wave was blocked by the LCT, which had not recovered sufficiently from previous activation. By the time the rightward-moving wave (shown by the 250 ms isochrone) reached the LCT it had recovered sufficiently and allowed the wave to cross the LCT into the pectinate muscle. The resulting idiopathic wave was a left-to-right plane wave (PW_{LR}) and it continued to circulate the model even as BCL was reduced to 370 ms.

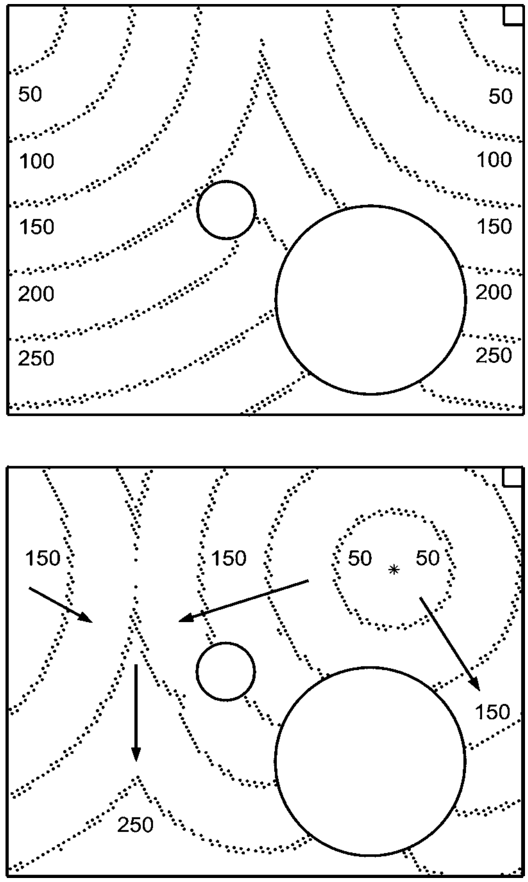

FIGURE 3. Activation maps. Upper panel: no pacing, isochrones are in milliseconds after SA node activated. Lower panel: pacing at BCL = 550 ms, stimulus site is marked with an asterisk, isochrones are in milliseconds after stimulus delivered.

Panel B of Fig. 4 shows the activation map of a single cycle of the PW_{LR} at a BCL of 370 ms. Zero time was arbitrarily chosen as the time when the wave spanned the CS. As the wave crossed the CS it split into two waves, a lower wave that moved toward and below the TCV and was blocked below the TCV and an upper wave that moved to the right into the pectinate muscle. At 100 ms, the upper wave was just to the left of the stimulus site when a stimulus was applied. The stimulus activated the tissue locally. The resulting stimulus wave propagated radially outward and collided with the PW_{LR} to the left of the stimulus site. The uncollided portions of the stimulus wave and the PW_{LR} merged (seen as a bulge in the 150 ms isochrone) and propagated to the right. The effect of the stimulus was to slightly alter the shape of the PW_{LR} around the stimulus site. Nevertheless, the PW_{LR} continued to circulate.

When the BCL was decreased to 360 ms, the stimulus came earlier so the PW_{LR} was farther to the left of the stimulus site when the stimulus was applied. This magnified the bulge in the PW_{LR}, advancing it and reducing its cycle time. Therefore, when the PW_{LR} next reached the LCT,

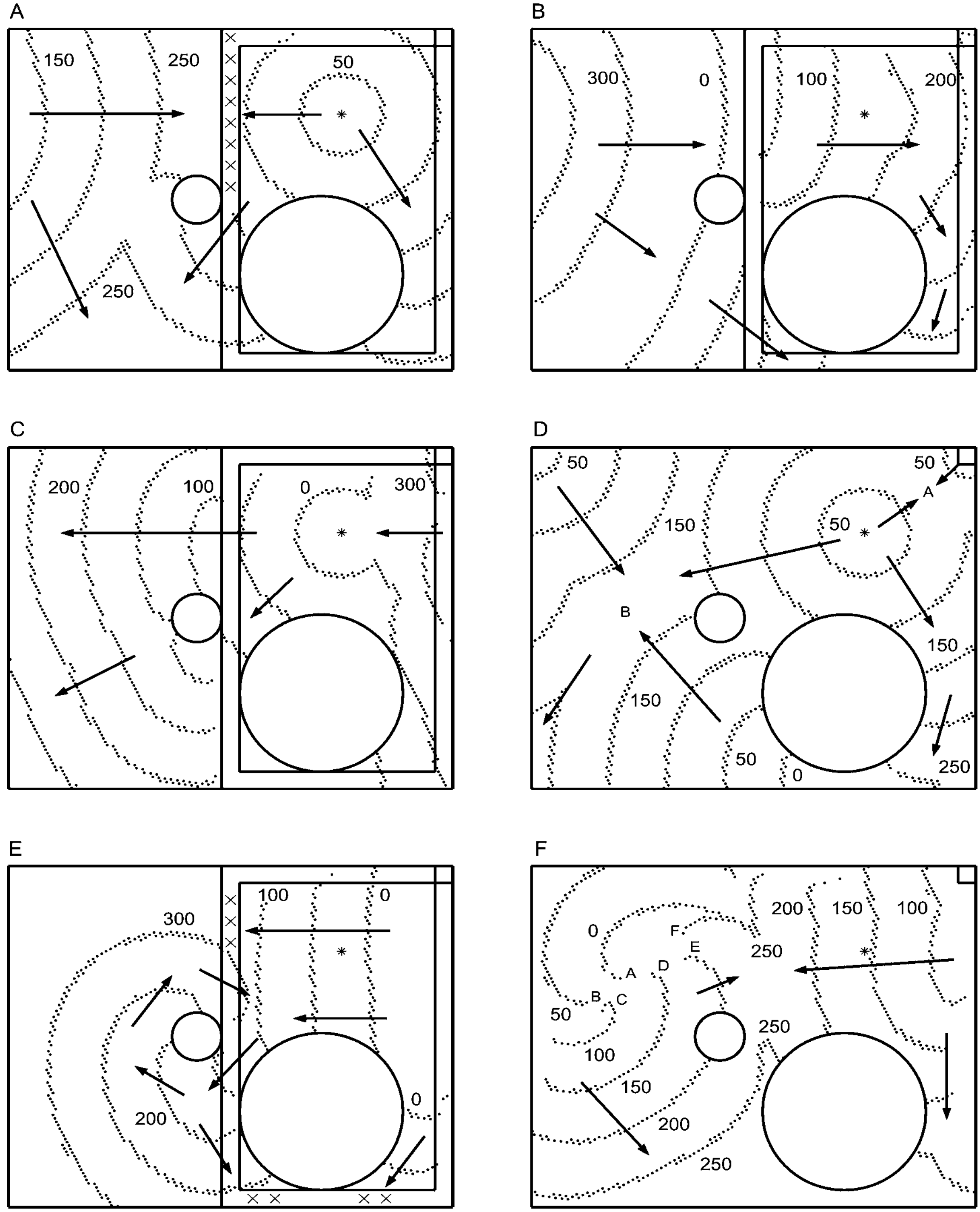

FIGURE 4. Examples of idiopathic waves: (A–B) left-to-right plane wave (PW_{LR}), (C) right-to-left plane wave (PW_{RL}), (D) anatomic reentry around TCV (ARW_{TCV}), (E) anatomic reentry around CS (ARW_{CS}), (F) spiral wave (SW). Stimulus site shown with asterisk. Isochrones in milliseconds after stimulus delivered (A, D) or after time 0 arbitrarily chosen (B, C, E, F). Location of block marked with X's. Crista terminalis not shown in Panels D and F for visual clarity. Detailed explanation of panels provided in text.

the LCT had not recovered sufficiently and the PW_{LR} was blocked and terminated. At the next stimulus, the response reverted to 1:1.

The response remained 1:1 for four stimuli but became increasingly irregular until a right-to-left plane wave (PW_{RL}) was generated at a BCL of 350 ms. This PW_{RL}, shown in Panel C of Fig. 4, persisted until BCL was decreased to 300 ms during which a stimulus wave was sufficiently large to collide with and extinguish the PW_{RL}. A second stimulus resulted in a stimulus wave that moved

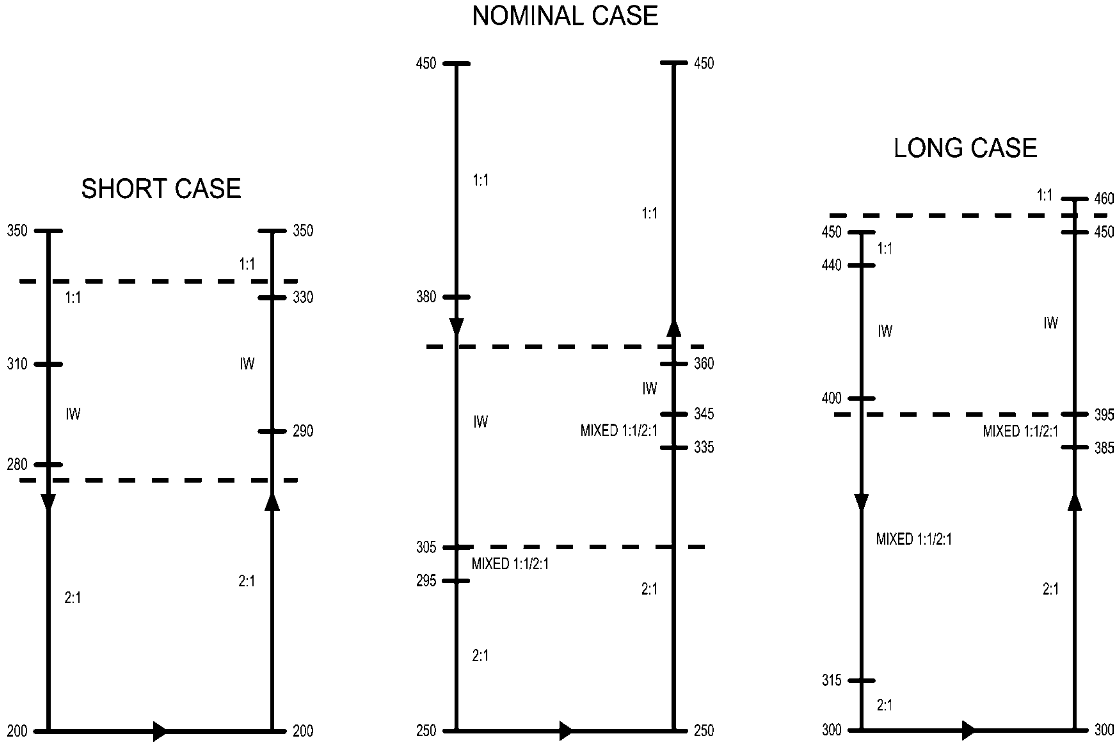

FIGURE 5. Global response diagrams for short case (left), nominal case (middle), and long case (right). In all panels, numbers outside are BCL (ms) and labels inside indicate the type of the response; IW denotes idiopathic waves. Dashed horizontal lines mark the global bistability window.

radially outward from the stimulus site. However, the wave was blocked on all sides by the CT. A third stimulus resulted in a stimulus wave that moved radially outward from the stimulus site that was not blocked by the CT. The resulting activation pattern was similar to the 1:1 pattern of the bottom panel of Fig. 3. This sequence repeated. In essence, the pectinate muscle, which was inside the CT, was responding 1:1 and the CT and non-pectinate muscle were responding 2:1. This mixed 1:1/2:1 response illustrates spatial bistability in that at the same BCL the model responded 1:1 in the pectinate muscle and 2:1 in the CT and non-pectinate muscle.

At a BCL of 290 ms, the response was 2:1 both inside and outside the CT. The effective BCL away from the stimulus site during 2:1, 580 ms, was twice the BCL at the stimulus site. Therefore, it was not surprising the activation pattern was similar to that obtained when pacing at a BCL of 550 ms, as shown in the bottom panel of Fig. 3.

As the BCL decreased to the end of the downsweep at 250 ms and then increased to 330 ms during the upsweep, a 2:1 response continued. At a BCL of 340 ms, the response was mixed 1:1/2:1 where the pectinate muscle was responding 1:1 and the CT and non-pectinate muscle were responding 2:1.

At a BCL of 350 ms the response was an anatomic reentrant wave (ARW_{TCV}) as shown in Panel D of Fig. 4. Zero time was chosen as the time when an activation wave was below the TCV. As this wave circulated about the TCV, the stimulus site and SA node activated roughly simultaneously. The stimulus wave and SA node wave collided and annihilated at position A. The stimulus wave, SA node wave, and the TCV wave collided and merged at position B and moved off the model. A portion of the stimulus wave moved downward along the right side of the TCV. This pattern repeated throughout pacing at a BCL of 350 ms.

At a BCL of 360 ms, the activation pattern of Panel D of Fig. 4 was broken when the TCV wave was blocked below the TCV. After one transition beat in which both the stimulus and the SA node activated and the subsequent waves collided and partially annihilated and/or merged, the response reverted to 1:1 where it remained for the remainder of the upsweep.

The response of the model is summarized in the center diagram of Fig. 5 which shows the transition points between the responses described above. The figure also shows that the downsweep/upsweep exhibited global 1:1/2:1 bistability covering a window of 60 ms and that idiopathic waves occurred at BCLs near and within the bistability window.

Short Case and Long Case

To investigate the effects of shifting the global bistability window to shorter (longer) BCLs, intrinsic APD across the model was decreased (increased) as shown in

TABLE 3. Prevalence of idiopathic waves.

Case	BCLs	Idiopathic	PW_{LR}	PW_{RL}	ARW_{CS}	ARW_{TCV}	SW
Short	16	6	4	0	3	0	2
Nominal	21	8	3	6	0	3	1
Long	17	6	5	4	5	1	5
All cases	54	20	12	10	8	4	8

Note. BCLs: Total number of BCLs in downsweep/upsweep. Idiopathic: Number of BCLs with idiopathic waves of any type observed. PW_{LR}: Number of BCLs with left-to-right plane waves observed. PW_{RL}: Number of BCLs with right-to-left plane waves observed. ARW_{CS}: Number of BCLs with anatomic reentrant waves around CS observed. ARW_{TCV}: Number of BCLs with anatomic reentrant waves around TCV observed. SW: Number of BCLs with spiral waves observed.

Table 2. A 350/200/350/10 downsweep/upsweep was applied to the model in the short case, and a 450/300/450/10 downsweep/upsweep was applied in the long case. The results, summarized in the left and right diagrams of Fig. 5, were qualitatively similar to those of the nominal case except that in the short case there was no mixed 1:1/2:1 response where the pectinate muscle responded 1:1 and the CT and non-pectinate muscle responded 2:1. Quantitative differences in ACL and the transition points between 1:1 and 2:1 responses are summarized in Table 2. In all three cases, the global bistability window was 60 ms.

Panel E of Fig. 4 illustrates an anatomic reentrant wave (ARW_{CS}), which occurred during pacing at a BCL of 400 ms on the downsweep (long case). A PW_{RL} moved across the pectinate muscle above the TCV and was partially blocked by the LCT. A portion of the wave was not blocked by the LCT and traveled between the CS and TCV. This wave remained anchored to and circulated twice about the CS thereby becoming an anatomic reentrant wave. At the end of the second revolution it was blocked by the LCT immediately next to the CS, became detached from the CS, traveled across the pectinate muscle and was extinguished by its own tail which had crossed the RCT.

Panel F of Fig. 4 illustrates a spiral wave (SW), which was initiated by an ARW_{CS} that became detached from the CS during pacing at a BCL of 450 ms on the upsweep. The letters A-B-C-D-E-F show the meandering of the spiral tip. This wave rotated for four cycles until the wave's tip collided with a portion of its tail that had separated. The response then reverted to 1:1.

Types and Prevalence of Idiopathic Waves

For all three cases, the observed idiopathic waves fell into five categories: left-to-right plane waves (PW_{LR}, see Fig. 4A-B), right-to-left plane waves (PW_{RL}, Fig. 4C), anatomic reentrant waves around the TCV (ARW_{TCV}, Fig. 4D), anatomic reentrant waves around the CS (ARW_{CS}, Fig. 4E), and spiral waves (SW, Fig. 4F). Though a given occurrence of a type of an idiopathic wave was not exactly the same as other occurrences, they were sufficiently similar to be grouped together and were clearly different than those of other categories.

Table 3 summarizes the prevalence of idiopathic waves in terms of the number of BCLs in which idiopathic waves were observed. Table 4 shows the number of each type of idiopathic wave and the average duration of each type of idiopathic wave in terms of the number of stimuli and amount of time during which each wave persisted. Both tables show that plane waves were more prevalent and lasted considerably longer than anatomic reentrant waves and spiral waves. While there were not necessarily more individual plane waves they were much more persistent because they had a convenient reentrant circuit bounded by the SVC and IVC that was sufficiently long relative to their activation wavelength to allow reentry. Their vertical extent was much larger than the anatomic obstacles and stimulus waves they encountered which allowed them to merge behind anatomic obstacles and merge with stimulus waves. Considering the plane waves, the PW_{LR}'s and PW_{RL}'s were roughly equally prevalent with the greater number of PW_{LR}'s being counteracted by the PW_{RL}'s lasting longer.

Anatomic reentrant waves had less room to operate than plane waves and therefore were more susceptible to being extinguished. Their reduced operating area was particularly severe between the CS and TCV and between the TCV and the bottom of the model. The presence of the CT with its

TABLE 4. Number and duration of idiopathic waves.

Idiopathic type	Number	AVG stimuli	AVG time (ms)
PW_{LR}	13	6.7	2333
PW_{RL}	4	16.0	5715
ARW_{CS}	10	2.4	863
ARW_{TCV}	8	1.0	360
SW	8	2.1	849

Note. Number: Number of individual idiopatic waves observed. AVG stimuli: Average number of stimuli during which an idiopathic wave persisted. AVG time: Average time during which an idiopathic wave persisted. This table combines results from all three cases (nominal, short, long). Abbreviations for idiopathic wave types are defined in Table 3.

longer intrinsic APD further increased the likelihood of conduction block in those regions. Similarly, spiral waves had less room to operate than plane waves. Furthermore, their cores were never stationary thereby increasing the likelihood of being extinguished by stimulus waves or moving into anatomic objects or off the top or bottom of the model. The anatomic reentrant waves and spiral waves occurred about the same number of times. However, ARW_{TCV}'s were shorter-lived which can be explained by the presence of the two restricted areas around the TCV.

Global and Local Dynamics

To assess whether complex dynamics near the stimulus site contributed to idiopathic waves, a sheet having uniform electrophysiologic properties of the pectinate muscle (site of stimulus) was exercised. Bifurcation diagrams of APD as a function of BCL for each of the three cases (short, nominal, and long) were generated and are shown in Fig. 6. The bifurcation diagrams and analysis of the supporting data for those diagrams confirms there was no complex dynamics in a pectinate muscle sheet. This suggests the idiopathic waves observed in the model were not caused by any unusual dynamics at or near the stimulus site which was in the pectinate muscle.

The next question is how the global dynamics of the entire model relates to the local dynamics of the pectinate muscle, non-pectinate muscle, and CT. To address that question, the global bistability window of the entire model and the local bistability windows of the three tissue types were compared.

To determine the local bistability windows, simulations were performed on one-dimensional fibers of Fig. 2A, corresponding to the pectinate and non-pectinate muscle and to CT. Bifurcation diagrams were generated for each fiber in each of the three cases (short, nominal, long) and are shown in Fig. 7. The two top panels of Fig. 7 show that in the nominal and long cases all three structures show 1:1/2:1 local bistability and there are no instances of a 2:2 response (alternans). The bottom panel of Fig. 7 shows that in the short case: 1) The pectinate and non-pectinate muscle show neither local bistability nor a 2:2 response. 2) The CT shows no 1:1/2:1 local bistability but shows 2:2/2:1 local bistability at 320 ms.

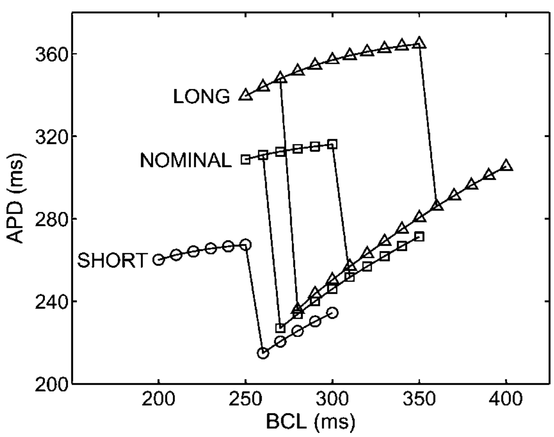

FIGURE 6. Local dynamics. Bifurcation diagrams obtained from pectinate muscle sheet (stimulus site) for short, nominal, and long cases.

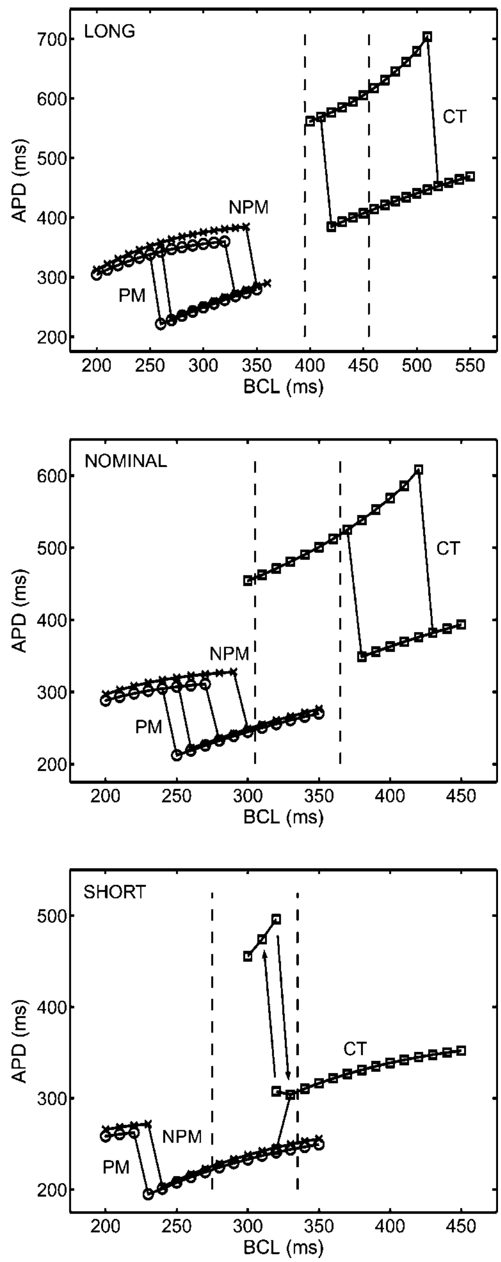

FIGURE 7. Bifurcation diagrams illustrating local dynamics for the long (top), nominal (middle), and short (bottom) case. The three diagrams shown in each panel correspond to non-pectinate muscle (NPM, *x*), pectinate muscle (PM, circles), and crista terminalis (CT, squares), and were obtained from one-dimensional fibers shown in Fig. 2A. Dashed lines show global bistability windows.

Eliminating Global Bistability and Effect on Idiopathic Waves

As shown in Figs. 6 and 7, shortening intrinsic APD by increasing τ_{slow} narrowed and eventually eliminated the local bistability window in the pectinate and non-pectinate muscle. Reducing intrinsic APD beyond that used in the short case was expected to eliminate bistability in not just the individual regions but the entire model as well. However, further increases in τ_{slow} relative to the short case did not reduce APD appreciably, so the elimination of the global bistability was effected by changing τ_{ung1} of the short case (Table 2).

With no pacing, the SA node activated at an ACL of 500 ms. In the same manner as the short case, a 350/200/350/10 downsweep/upsweep was applied. There was no global 1:1/2:1 bistability. On both the downsweep and upsweep the response was 1:1 at BCLs of 320 ms and longer and 2:1 with intermittent activation of the SA node at BCLs of 310 ms and shorter. There were no sustained idiopathic waves at any BCLs. The only irregular behavior observed was four single ARW_{TCV}'s, each of which lasted for one beat or less and occurred at different BCLs. This result suggests that global 1:1/2:1 bistability is required for the existence of idiopathic waves during a downsweep/upsweep.

DISCUSSION

Comparison to Experiments

This study used the computer model to interpret experiments performed on sheep atria *in vivo*, whose results first indicated the correlation between the global 1:1/2:1 bistability and arrhythmogenesis.[16] In this experiment, a downsweep/upsweep protocol was applied to the free wall of the sheep right atrium. Bistability was observed in 17 of 19 trials in 5 of 6 sheep, and the idiopatic waves occurred at BCLs within the bistability window or near its limits. In the model presented here the location of the pacing site and the details of the pacing protocol were based on the sheep experiment. In doing so, the model was successful in reproducing the key observation that the idiopatic waves occurred within the global 1:1/2:1 bistability window or near its limits.

In contrast to the experiment, in which only a 2×2 cm region of the free wall was mapped, the model monitored electrical activity over the entire atrium, allowing us to elucidate the nature and the mechanisms of the idiopatic waves. Five major types of idiopathic waves were identified (Fig. 4) and their prevalence and persistence were quantified (Tables 3 and 4). Because of the different field of view, it is not possible to compare directly the types of idiopathic waves seen in the model and experiment: in a small portion of the atrium mapped, the experiment could only distinguish between single, multiple, and rotor-like waves while the model could distinguish five types of waves based on their mechanisms (plane waves, reentries, and spiral waves). Nevertheless, the data on prevalence are consistent: single waves were most prevalent and rotor-like waves were least prevalent in the experiment, while plane waves were most prevalent and the spiral waves were least prevalent in the model (Table 3).

Involvement of Model Components in Idiopatic Waves

A more comprehensive analysis afforded by the model allowed us to investigate the role of the stimulus and the anatomic structures in the initiation and termination of idiopatic waves (examples shown in Fig. 4 and discussed in text). While there was a great number and variety of specific mechanisms, some trends emerged. These trends are summarized in Table 5, which lists the number of occurrences of each type of idiopathic wave and the number of times the stimulus and right atrial structures were involved in their initiation and/or termination by blocking waves, causing wave breaks, launching waves, etc.

CT

The CT was involved in 40 of the 43 idiopathic waves. By having the longest intrinsic APD the CT was more likely to block waves than the adjoining non-pectinate muscle and pectinate muscle. Of course, based on the timing of previous activations, each portion of the CT did not block in unison. In fact, often even a small section would not block in unison or to the same degree. The resulting selective blocking by the CT led to wavebreaks, changes in direction of wave propagation and most significantly caused and terminated idiopathic waves. Specific examples of CT involvement are shown in Panels A and E of Fig. 4.

Stimuli

Stimuli were involved in 33 of the 43 idiopathic waves. Such frequent involvement is not surprising given the stimuli were constantly being applied. The stimuli were

TABLE 5. Role of model components.

Idiopathic type	Number	CT	STIM	CS	TCV	SA
PW_{LR}	13	13	11	3	0	1
PW_{RL}	4	3	3	0	0	1
ARW_{CS}	10	10	8	10	3	1
ARW_{TCV}	8	8	8	0	8	1
SW	8	6	3	1	2	0
Total	43	40	33	14	13	4

Note. Number: Number of individual idiopatic waves observed. CT (STIM, CS, TCV, SA): Number of idiopatic waves in which CT (stimulus, CS, TCV, SA node) were involved. This table combines results from all three cases (nominal, short, long). Abbreviations for idiopathic wave types are defined in Table 3.

often the source for idiopathic waves. Specifically, instead of propagating as in a normal 1:1 response, stimulus waves would be broken, blocked, and/or directed in abnormal ways based on preceding activation patterns and/or anatomic structures. Stimuli were also involved in terminating idiopathic waves when stimulus waves collided with idiopathic waves. Finally, though not shown in Table 5, stimuli often affected existing idiopathic waves when stimulus waves would collide and merge with idiopathic waves or collide and alter the direction of idiopathic waves. In some cases, the collision and merger of stimulus waves and idiopathic waves would result in entrainment. Specific examples of stimuli involvement are shown in Panels A and D of Fig. 4.

Anatomic Openings

Of the 43 idiopathic waves, the CS was involved in 14 and the TCV was involved in 13. Their involvement was expected as they provided natural anchoring structures for reentrant waves, the CS was centrally located, and the TCV occupied a large section of the model. Specific examples of CS and TCV involvement are shown in Panels D and E of Fig. 4.

SA Node

The SA node was directly involved in only 4 of the 43 idiopathic waves. Such a low number is attributable to the SA node activating infrequently since it was reset by the stimulus waves and idiopathic waves that crossed it. It is important to note that this observation is somewhat misleading: a simulation in which the SA node was removed failed to produce idiopathic waves. Thus, even though the SA node may not been directly involved with individual idiopathic waves, these results suggest it was important to the formation of idiopathic waves overall. Though infrequently, the SA node activated to varying degrees in each case. Such activation would launch waves that would alter the condition of recovery in areas crossed by the waves thereby increasing the dispersion of refractoriness and reducing the overall stability of the model.

Global Versus Local Dynamics in Initiation of Idiopathic Waves

Traditionally, the onset of arrhythmias has been attributed to a conduction block caused by heterogeneity in the electrophysiological properties of the tissue.[26] Findings of this study fit into this general framework: as discussed above, conduction block caused by CT most frequently contributed to idiopatic waves. However, conduction blocks can occur owing to different mechanisms. This study identified global 1:1/2:1 bistability as a cause. The causal relationship between global bistability and idiopatic waves is supported by two observations: 1) shifting the global bistability window results in a corresponding shift in the idiopathic waves and 2) eliminating global bistability window prevents formation of idiopatic waves. Of course, there exist other mechanisms that can lead to arrhythmia, such as 1:1/2:2 bistability[25] or discordant alternans.[21] However, the 1:1/2:1 bistability explored here may be most important as it is more prevalent than other dynamic responses.[8]

What are the conditions required for global 1:1/2:1 bistability to arise and how do they relate to the local dynamics of the underlying tissue? Figure 7 shows the local bifurcation diagrams of the pectinate muscle, non-pectinate muscle, and CT together with the global bistability windows in all three cases (short, nominal, long). It suggests that the global bistability windows are correlated more closely with the local dynamics in the CT than in the pectinate muscle and non-pectinate muscle. This is consistent with the CT having a longer intrinsic APD and therefore transitioning from 1:1 to 2:1 at longer BCLs. In that sense, the CT "controls" global dynamics and the local dynamics of the pectinate muscle and non-pectinate muscle are less significant.

The bottom panel of Fig. 7 is particularly interesting as neither the pectinate muscle nor non-pectinate muscle show local bistability yet there is global bistability. This result suggests that local bistability in those structures is not necessary for global bistability. Also, given that global bistability occurred regardless of whether or not each of those structures showed local bistability further supports the observation that the local dynamics in those structures is less significant. The CT shows local bistability only at 320 ms and that bistability is 2:2/2:1 as opposed to the 1:1/2:1 bistability seen in the whole model. Such a narrow bistability window suggests that local bistability in the CT may not be necessary for global bistability. It appears that global bistability requires that local transitions from 1:1 to 2:1 and from 2:1 to 1:1 occur at different BCLs but may not require local bistability anywhere. Therefore, the results of this study make it possible to clarify the general notion of electrophysiological heterogeneity that is responsible for the onset of arrhythmias: the underlying cardiac structures should transition from 1:1 to 2:1 response at different BCLs.

Limitations and Future Work

The idealized right atrium model used in this study provided considerable insight into the findings of the sheep experiments that originally revealed idiopathic waves during downsweep/upsweeps.[16] Nevertheless, the model had its own limitations. First, an idealized membrane model rather than a complex electrophysiologic model was used, and thus effects of the various ionic, pump, and exchange currents and ionic concentrations were not examined. Second, electrophysiologic inhomogeneity was limited to making discrete regions have different intrinsic APDs which may have exaggerated APD gradients between regions but underestimated overall inhomogeneity. Third, resistivity inhomogeneity and anisotropy were not modeled which is expected to impact the fine details of the observed waves. Fourth, the specific structures such as the SVC, IVC, CS, TCV, CT, SA node, pectinate muscle and non-pectinate muscle were only approximated. Fifth, the model did not include the effects of the left atrium, which is electrically connected to the right atrium and has been shown experimentally to play a role in atrial arrhythmias. Finally, the complex three-dimensional structure of the pectinate muscle was not modeled. However, the results of this study rely mainly on the anatomic structures and the heterogeneity of membrane properties. Because both are present in real atria and whole hearts, the following conclusions, drawn from this simulation study, are expected to be applicable to real atria and probably to whole hearts.

In this study, the model was also constrained by the goal of reproducing the experimental setup and pacing protocol. Thus, several questions had to be left unanswered. For example, in the downsweep/upsweep protocol, the tissue was being paced continuously. Thus, we do not know whether the observed idiopatic waves were self-sustained or whether they relied on the stimuli to persist. One could perform a simulation in which pacing would be stopped after idiopatic waves were detected, so that the spontaneous electrical activity could be examined. Likewise, one could investigate the sensitivity of the idiopatic waves formation to the pacing site and the details of pacing protocol, such as step size and number of paces at each BCL. However, we have decided that such simulations are beyond the scope of the present report and should be pursued as a separate study, perhaps in conjunction with animal experiments. Thus, with the incorporation of realistic pacing protocols, the cardiac model presented here has the ability to add to the body of knowledge concerning cardiac dynamics and arrhythmias, and it can be used to assess the effects of novel electrical and pharmacologic treatments of atrial arrhythmias.

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